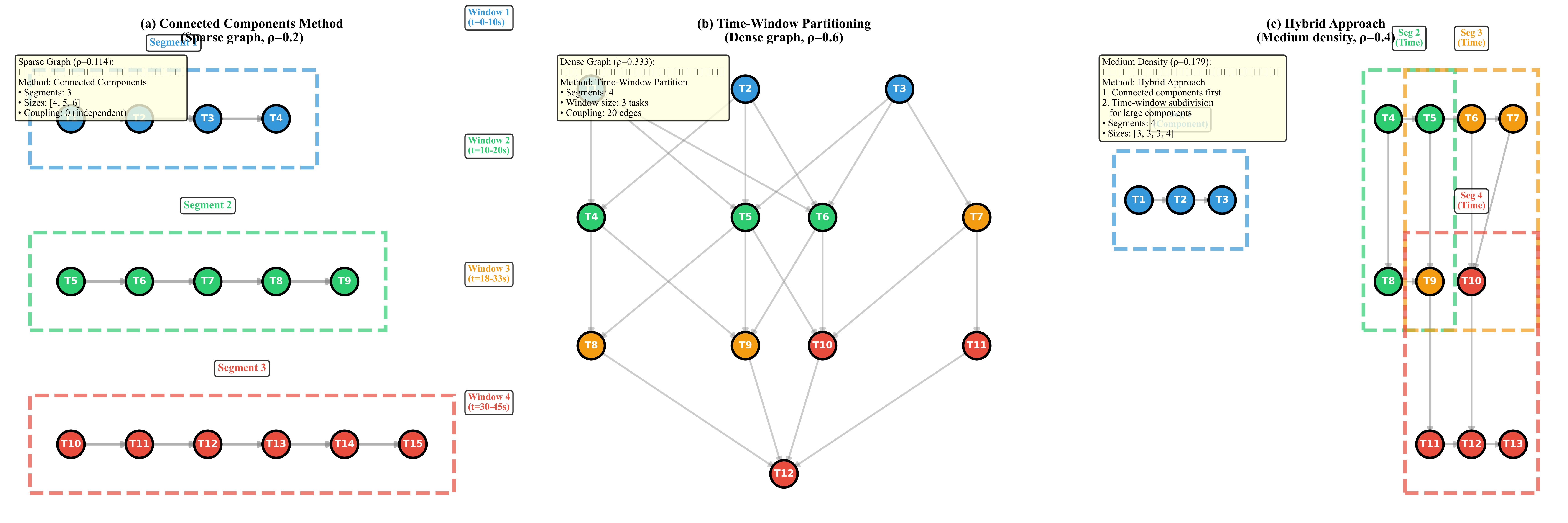

(a) Connected Components Method
(Sparse graph, ρ=0.2)
Segment 1
Sparse Graph (ρ=0.114):
Method: Connected Components
• Segments: 3
• Sizes: [4, 5, 6]
• Coupling: 0 (independent)
T2
T3
T4
Segment 2
T5
T6
T7
T8
T9
Segment 3
T10
T11
T12
T13
T14
T15
Window 1
(t=0-10s)
Window 2
(t=10-20s)
Window 3
(t=18-33s)
Window 4
(t=30-45s)
(b) Time-Window Partitioning
(Dense graph, ρ=0.6)
Dense Graph (ρ=0.333):
Method: Time-Window Partition
• Segments: 4
• Window size: 3 tasks
• Coupling: 20 edges
T2
T3
T4
T5
T6
T7
T8
T9
T10
T11
T12
(c) Hybrid Approach
(Medium density, ρ=0.4)
Medium Density (ρ=0.179):
Method: Hybrid Approach
1. Connected components first
2. Time-window subdivision
for large components
• Segments: 4
• Sizes: [3, 3, 3, 4]
Component)
T1
T2
T3
Seg 2
(Time)
Seg 3
(Time)
Seg 4
(Time)
T4
T5
T6
T7
T8
T9
T10
T11
T12
T13